Exc No: 01
27/7/26

Basic Linux Commands

Aim: To study various network commands used in Linux and Windows.

1) arp -a : IP address of your computer

O/P:

Internet Address	Physical Address	Type
192.168.1.1	44-95-3b-a6-8a-30	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static

2) hostname : Displays the computer name

O/P: LAPTOP-4DBQ1LHG

3) ipconfig /all : Displays complete IP configuration.

O/P: Window IP Configuration

Host Name : LAPTOP-4DBQ1LHG
Primary Dns Suffix :
Node Type : Mixed
IP Routing Enabled : No
WINS Proxy Enabled : No

4) nbtstat -a : Displays NetBIOS information of a remote computer

O/P: NBTSTAT [[-a RemoteName] [-A IP Address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

5) netstat : Display active network connections and ports

O/P: Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:1072	Kubernetes :1073	Established
TCP	127.0.0.1:1073	Kubernetes :1072	Established

6) nslookup : Finds the IP address of a website

nslookup google.com

O/P: Server : Unknown
Address : 192.168.1.1

7) pathping : Shows the path and packet loss to a destination.

pathping ~~google~~.com

O/P: Tracing route to google.com [142.250.182.78]
over a maximum of 30 hops:
0 LAPTOP-4DBQ1LHG [192.168.1.9]
1 192.168.1.1

8) ping : Checks whether another device is reachable.

ping google.com

O/P: Reply from 142.251.43.46 : bytes = 32 time = 4ms TTL = 119
Reply from 142.251.43.46 : bytes = 32 time = 8ms TTL = 119

9) Route : Displays the routing table.

route print

O/P: Active Routes :

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.9	45
127.0.0.0	255.0.0.0	on-link	127.0.0.1	331

10) ip

ip <options> <object> <command>

a) To show the IP address assigned :

ip address show

O/P: enp0s3 :

inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3 valid_lft 85092sec preferred_lft 14297 sec

b) To assign an IP :

ip address add 192.168.1.254/24 dev enp0s3

O/P: enp0s3 :

inet6 fe80::82bf:68de:84fd:5451/64 scope link noprefixroute valid_lft forever preferred_lft forever

c) To delete an IP :

ip address del 192.168.1.254/24 dev ep POS3

O/P: inet6 fd17:625c:f037:2:42cd:cd59:ab53:d22
/64 scope global dynamic noprefixroute valid_lft
86175 sec preferred_lft 14175 sec

d) Alter the status of the Interface:

i) ip link set eth0 up

O/P: mtu 1500 qdisc fq_codel state UP group default
qlen 1000 altname

ii) ip link set eth0 down

O/P: mtu 1500 qdisc fq_codel state DOWN group
default qlen 1000 altname

iii) ip link set eth0 promisc on

O/P: <PROMISC> mtu 1500 qdisc fq_codel state DOWN
group default qlen 1000 altname

e) Add Default Route via local gateway:

ip route add default via 192.168.1.254 dev eth0

O/P: eth0 : mtu 1500 qdisc fq_codel state UP group
default qlen 1000

f) Add the route

i) Via gateway : ip route add 192.168.1.0/24 via 192.168.1.254

O/P: 192.168.1.0/24 via 192.168.1.254 dev eth0 proto static

ii) Can be reached on device : ip route add 192.168.1.0/24 dev eth0

O/P: change the valid_lft & preferred_lft

g) Delete the route

i) Via gateway : ip route delete 192.168.1.0/24 via 192.168.1.254

O/P: Remove the ip from ip address show.

ii) Display the route : ip route get 10.10.1.4

O/P: 10.10.1.4 via 192.168.1.254 dev eth0 src 10.0.2.15 uid 0 cache

11) ifconfig : It is a staple in many sysadmin's tool belt for configuring and troubleshooting networks.

O/P: eth0 : flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500 inet 10.0.2.15

lo : flags=73<UP,LOOPBACK,RUNNING> mtu 65536 inet 127.0.0.1

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12) mtr :

mtr <options> hostname/IP

a) Shows statistics with time and loss %.

mtr google.com

	Host	Loss%	Snt	Last	Avg	Best	Wrst	StDev
O/P:	pnmaaa-au-in-f14.1e100.net	0.0%	54	16.7	8.8	3.2	60.2	11.3

b) Show numeric IP address

mtr -g google.com

O/P: google.com (142.251.221.174)

c) Show numeric IP address and hostnames

mtr -b google.com

O/P: 142.251.221.174 , pnmaaa-au-in-f14.1e100.net

d) Set numbers of pings

mtr -c 10 google.com

	Host	Loss%	Snt	Last	Avg	Best	Wrst
O/P:	pnmaaa-au-in-f14.1e100.net	0.0%	54	8.8	6.1	3.2	36.2

13) tcpdump

i) Install: dnf install -y tcpdump

ii) tcpdump -D

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iii) tcpdump -i eth0 : Capture traffic on eth0

O/P: 20 packets captured, 20 packets received by filter, 0 packets dropped by kernel

iv) tcpdump -i eth0 -c 10 : Capture traffic to and from one host

O/P: 10 packets captured, 10 packets received by filter, 0 packets dropped by kernel

v) To find traffic coming from and going to 8.8.8.8

tcpdump -i eth0 -c 10 host 8.8.8.8

O/P: 5 packets captured, 5 packets received by filter

vi) Traffic coming 8.8.8.8

tcpdump -i eth0 src host 8.8.8.8

O/P: 3 packets captured, 3 packets received by filter

vii) Outbound traffic going to 8.8.8.8:

tcpdump -i eth0 dst host 8.8.8.8

O/P: 2 packets captured, 2 packets received by filter

viii) Capture traffic to and from a network

tcpdump -i eth0 net 10.1.0.0/24

O/P: 0 packets captured, 0 packets received by filter

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ix) Capture traffic to and from port numbers

tcpdump -i eth0 port 53

O/P: 10 packets captured, 10 packets received by filter

x) specific host

tcpdump -i eth0 host 8.8.8.8 and port 53

O/P: 5 packets captured, 5 packets received by filter

xi) To capture only HTTPS traffic:

tcpdump -i eth0 -c 10 host www.google.com and port 443

O/P: 3 packets captured, 3 packets received by filter

xii) To capture all port except port 53 and 25

tcpdump -i eth0 port not 53 and not 25

O/P: 2 packets captured, 2 packets received by filter

14) nmcli

a) list NetworkManager connection

nmcli connection show

O/P:

NAME	UUID	TYPE	Device
Wired connection 1	a5eb6490-cc20-3668-81f8-0314a27f3f75	ethernet	enp0s3

b) Rename the connection profile:

nmcli connection modify "Wired connection 1"

c) Configure IPV4 settings:

i) Use DMCD,

nmcli connection modify "Wired connection 1" ipv4.method auto

O/P: ipv4.method : auto

ii) Set static IPV4 address

nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.address 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com

O/P: Modify the IP address.

d) Active the profile

nmcli connection up Internal-LAN

O/P: Connection successfully activated.

e) Display the IPV4 default: ip route show

O/P: default via 192.168.1.254 dev eth0

f) Display the IPv6 default: ip -6 route show

O/P: fd17:625c:f037:2::/64 dev eth0 proto ra metric 100 pref medium

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Student observation:

1) Which command is used to find the reachability of a host machine from your device?

Ans: Ping

Syntax: Ping <hostname>

2) Which command will be give the details of hops taken by a packet to reach its destination?

Ans: traceroute

3) Which commands displays the ip configuration of your machine.

Ans: ipconfig or ifconfig

4) Which command displays the TCP port status in your machine?

Ans: netstat

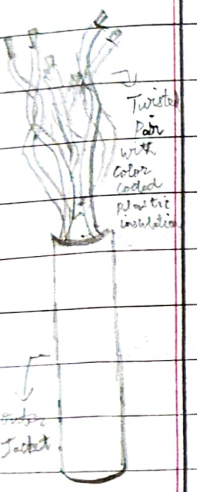
5) Write the command to modify the IP configuration in a linux machine.

Ans. ip address add 192.168.1.254/24 dev eth0s3

Ex No: 02
28/7/26 Aim: Study of different types of Network cables.

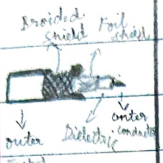
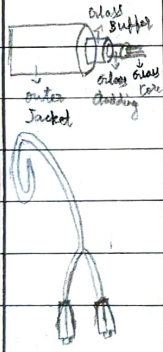
a) Understand different types of network cable.

1) Unshielded Twisted Pair (UTP) Cable
2) Shielded Twisted Pair (STP) Cable
3) Coaxial Cable
4) Fibre Optic Cable





Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **Disadvantages** • More prone to (EMI) Electromagnetic Interference and noise	10 Base-T Ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	**Advantages** • Shielded • Faster than UTP • Less susceptible to noise and interference. **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet 10G Ethernet (55 m) widely used in data centres
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)

Coaxial cable 	RG-6 RG-59 RG-11	10-100 Mbps	Advantages • High bandwidth • Immune to Interference • Low loss bandwidth • Versatile Disadvantages • Limited distance • Cost • Size is bulky	Speed of signal is 500 m. Television network High speed internet connection
Fibre optics cable 	Single mode Multi mode	100 Gbps	Advantages • High Speed • High Bandwidth • High security • Long distance Disadvantages • Expensive • Requires skilled installer	Maximum distance of fibre optics cable is around 100 metres

b) Make your own Ethernet Cross-over cable / Straight cable

Tools and parts needed:

- Ethernet cabling. CAT 5e is ertified for gigabet support, but CAT5 cabling works as well, just over shorter distance
- A crimping tool. This is an all-in-one networking

tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.

- Two RJ45 plugs
- optional two plug shields

Step 1: To start conversation of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than objects, snip them back to create an even level.

Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means the you stripped off too much shielding.

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeatly for the other end using the

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Student observation:

1) Difference between cross and straight cables?

Ans: Straight Cable : Same pin standards on both ends; connects difference device type

Cross Cable : Swapped pin standards on ends; connects identical device types.

2) Cable to connect two PC's?

Ans: Cross table

3) Cable to connect a router / switch to a PC?

Ans: Switch to PC : Straight Cable
Router to PC : Cross Cable

4) Category for twisted pair cable used in your lab?

Ans: CAT6

5) Understanding, challenges, and output of making a cable

Ans: * Understanding: Wires must match specific color standards to transmit data.

* Challenges: Lining up the tiny wires and keeping them in order inside the RJ45 plug.

* Output: A working cable that passing a LAN tester checker.

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Ex No : 03
3/08/26

Aim : To study the Packet tracker tool Installation and user Interface overview

c) To understand environment of CISCO Packet Tracker to design simple network

INTRODUCTION :

A simulator, as the name suggests, simulates network devices and its environment. Packet Tracker is an exciting network design, simulation and modelling tool.

1) It allows you to model complex systems without the need for dedicated equipment.

2) It helps you to practice your network configuration and troubleshooting skills via computer or an Andriod or iOS based mobile devices.

3) It is available for both the Linux and Windows desktop environments.

4) Protocols in Packet Tracker are coded to work and behave in the same way as they would be real hardware.

User Interface overview :

1) Menus bar : This is a common menu found in all software applications, it is used to open, save, print, change preferences and so on.

2) Workspace : This is the area where topologies are created and simulations are displayed.

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3) Main toolbar: This bar provides shortcut icons to many options that are commonly accessed such as open, save, zoom, undo and redo and on the right-hand side is icon for entering network information for the current

4) Logical / Physical Workspace tab: These tabs allows to toggle between the logical and physical work

5) Common tool bar: This toolbar provides controls for manipulating topologies, such as select, move layout, place note, delete, inspect, resize shape, and add complex PDU.

6) Real-time / Simulation tabs: These tabs are used toggle between the real and simulation modes. But are also provided to control the time, and to capt the packets.

7) Networking Component box: This component contains all the network and end devices available with Packet Tracker, and is further divided into two areas: Area 7a & Area 7b.

8) User-Created packet box: Users can create highly customized packets to test their topology from this area.

d) Analyse the behaviour of network devices using CISC

1) From the network component box, click and drag and-drop the below components:

a. 4 Generic PCs and One Hub
b. 4 Generic PCs and one Switch

2) Click on Connections :

a. Click on Copper Straight - Through cable.
b. Select one of the PC and connect it on HUB using the Cable. The link LED should glow in green, indicating that link is up. Similarly connect remaining 3 PCs to the HUB.
c. Similarly connect 4 PCs to the switch using copper straight - through cable.

3) Click on the PCs connected to hub, go to the Desktop tab, click on IP Configuration, and enter an IP address and subnet mask. Here, the default gateway and DNS server information is not needed as there are only two end devices in the network.

4) Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.

5) Repeat step 3 to step 5 for the PCs connected to the switch.

6) Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviours of switch & HUB.

Student Observation :

a) Packet Forwarding Behavior

HUB (broadcasting)

- Operates at Layer 1 (Physical)
- Cannot read MAC addresses.

- Replicates packets to all ports.
- Causes high data collisions
- Lacks privacy and efficiency

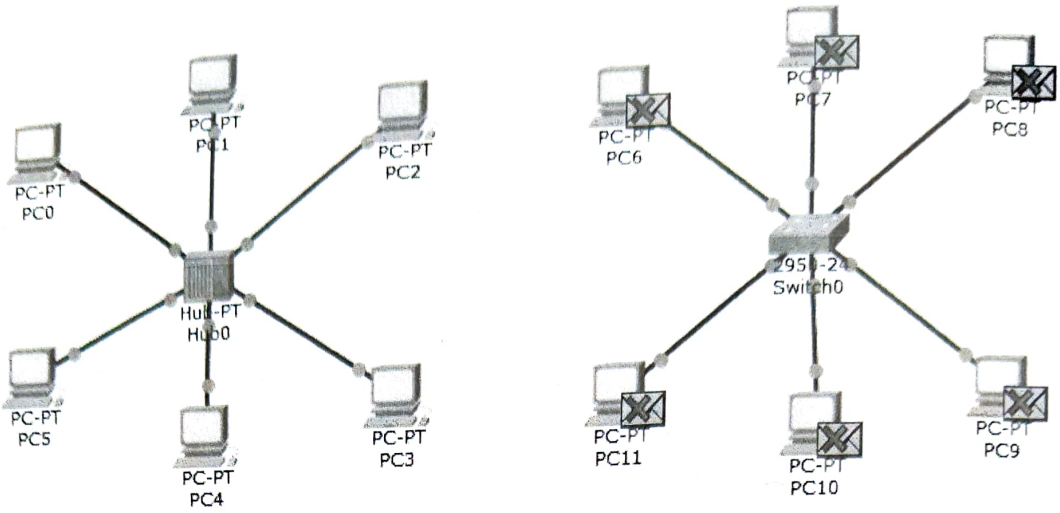
SWITCH (Unicasting)

- Operates at Layer 2 (Data Link)
- Reads destination MAC addresses.
- Forwards packets to one specific port.
- Eliminates network data collisions
- Greatly improves network security.

b) College Network Topology

Extended Tree Topology

- Internet Cloud : The main external source.
- Core Router : The central campus gateway.
- Distribution switches : Main building hubs.
- Access Switches : Individual classroom connections
- End Devices : Computers, Laptops and printers



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type	Info
	18.019	Switch0	PC9	STP	
	18.019	Switch0	PC10	STP	
	20.022	--	Switch0	STP	
	20.023	Switch0	PC6	STP	
	20.023	Switch0	PC8	STP	
	20.023	Switch0	PC11	STP	
	20.023	Switch0	PC7	STP	
	20.023	Switch0	PC9	STP	
	20.023	Switch0	PC10	STP	

Reset Simulation Constant Delay Captured to: 20.023 s

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 10.1.1.5

Pinging 10.1.1.5 with 32 bytes of data:

Reply from 10.1.1.5: bytes=32 time=1ms TTL=128
Reply from 10.1.1.5: bytes=32 time=4ms TTL=128
Reply from 10.1.1.5: bytes=32 time=3ms TTL=128
Reply from 10.1.1.5: bytes=32 time=3ms TTL=128

Ping statistics for 10.1.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

PC>
```

Ex No: 04
10/8/26

Setting Up a Local Area Network (LAN)

Aim:

To design, set up, and configure a local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network.

Equipment Required:

S. No	Equipment	Quantity	Specification
1	Personal Computer (PC)	4	Ethernet Enabled
2	Network switch	1	8/16/24 port, unmanaged/managed
3	Ethernet Cables (CAT5e/CAT6)	4	Straight-through, RJ45 connectors

Theory:

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or laboratory, allowing them to share resources such as files, printers, and internet connections.

The Star topology is used in this experiment, where all the end devices (PCs) are individually connected to a central network switch. The switch operates at Layer 2 of the OSI Level and forwards data frames between connected devices based on their MAC addresses

Implementation in Cisco Packet Tracer:

Step 1: Place the Devices

1) Open Cisco Packet Tracer
2) From the devices category panel, select End Devices and drag 4 PCs onto the workspace.
3) Select switches and drag 1 switch onto workspace.

Step 2: Cable the Devices

4) Click Connections → Select copper Straight-Through Cable
5) Click each PC, then click the switch, and select an available FastEthernet port to complete each connection.
6) Repeat for all four PCs. Green links lights confirms a successful physical connection.

Step 3: Assign IP addresses to PCs

7) Click a PC → Desktop tab → IP Configuration.
8) Enter the static IP address and Subnet Mask as per the IP table for each PC.

Step 4: Configure the Switch Management IP

9) Click the switch → CLI Tab

Step 5: Verify Connectivity Using Ping

10) Click a PC → Desktop tab → Command Prompt
11) Use the ping command to test connectivity
12) Successfully "Reply From ..." message confirm Connectivity

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Step 6: Demonstrate Data Sharing (Server Simulation)

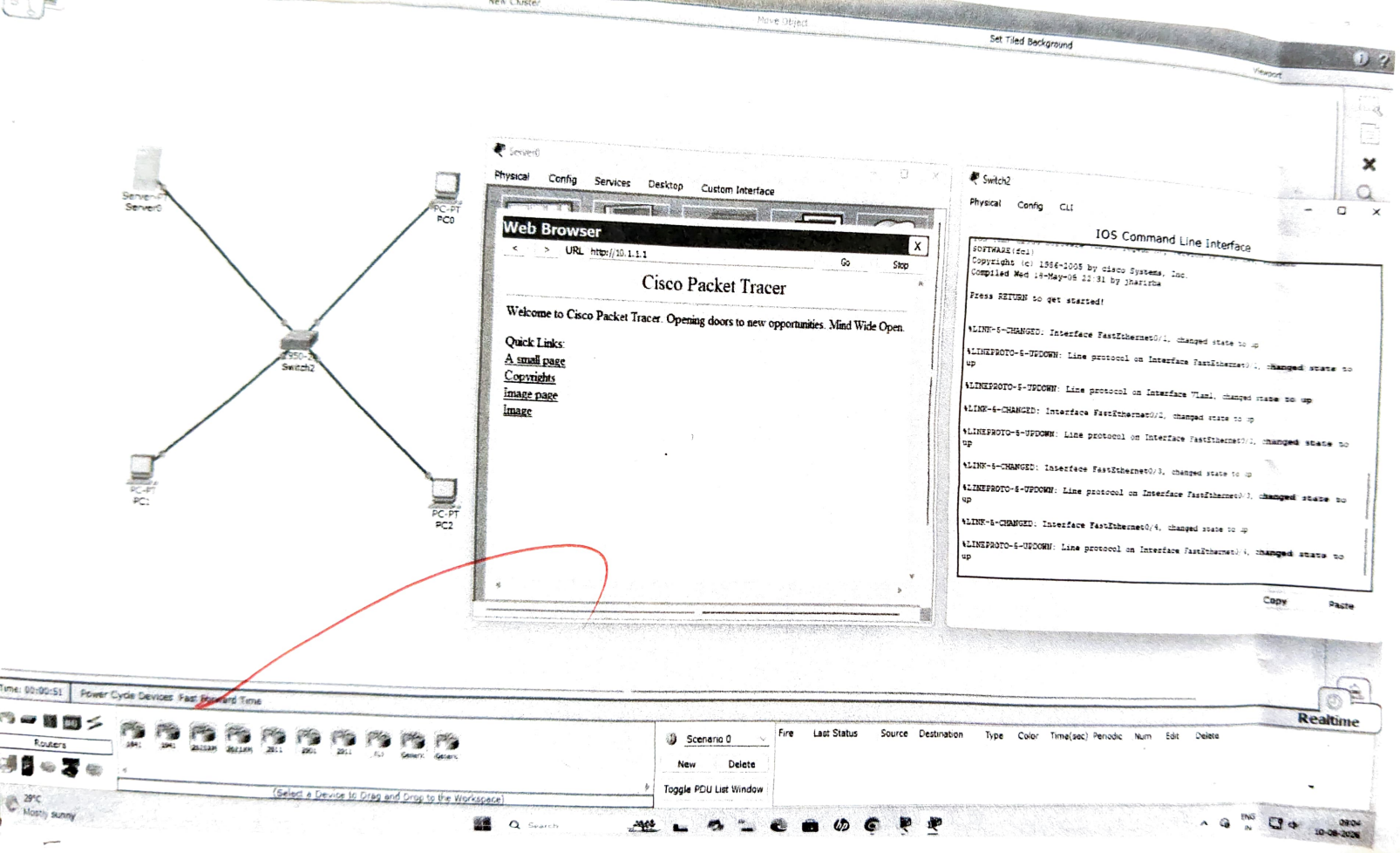
13) Add a server device and connect it to the switch
14) Assign it an IP address via its Desktop
15) Open the server's Services tab and enable HTTP service

Step 7: Observe Packet Flow in Simulation Mode

16) Switch to Simulation mode.
17) Send a ping or web request between two devices and use the Play / capture controls to step through the packet.

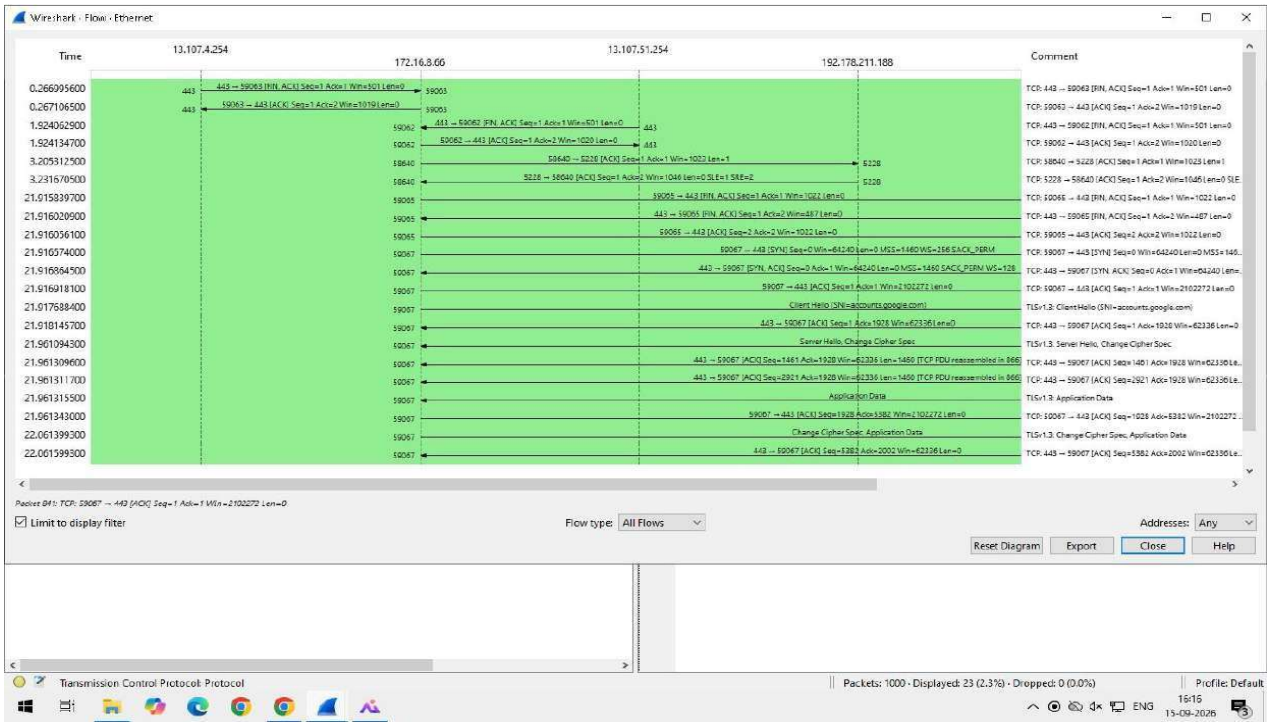
Observation:

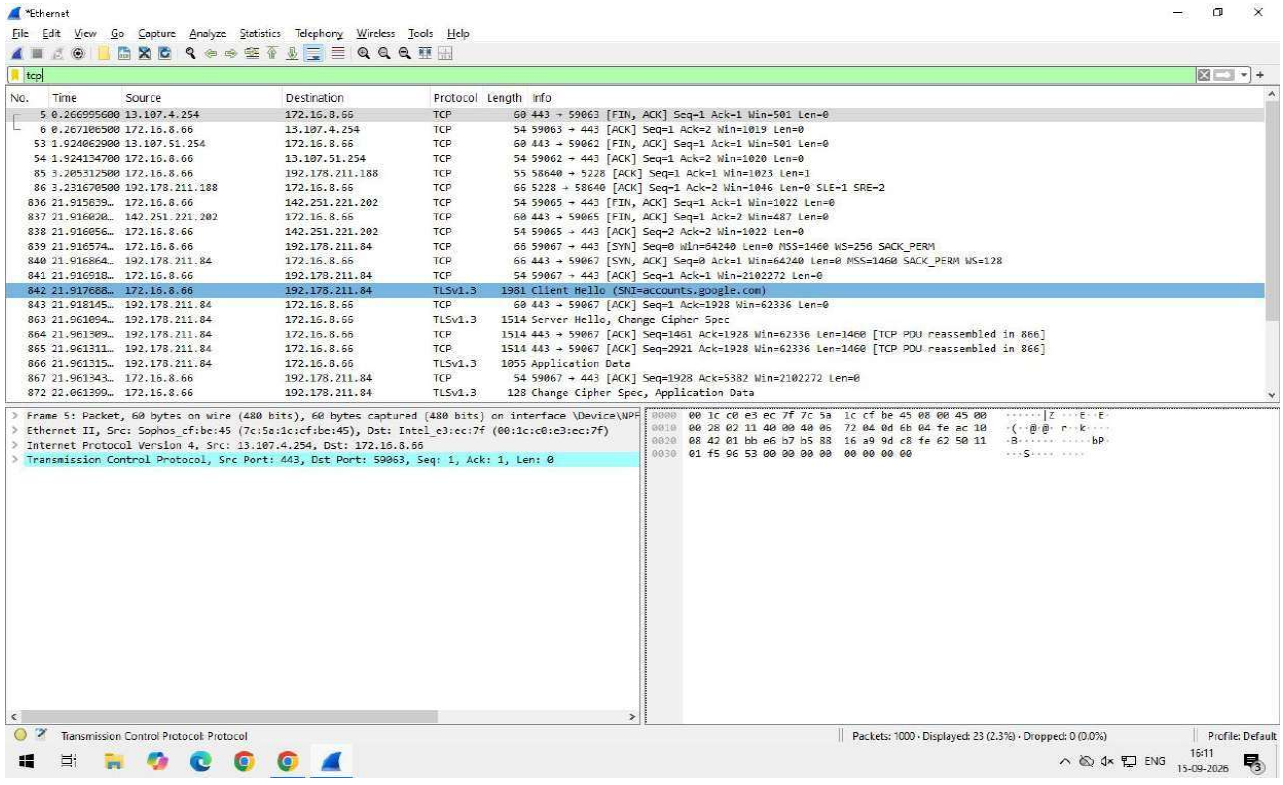
All four PCs were successfully connected to the switch in the star topology and assigned unique static IP addresses within the same subnet. Ping tests between all devices returned successful replies, confirming end-to-end connectivity


Server0
PC-PT PC0
2950-24 Switch2
PC-PT PC1
PC-PT PC2
Physical Config Services Desktop Custom Interface
Web Browser
URL http://10.1.1.1
Go Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
image page
Image
Switch2
Physical Config CLI
IOS Command Line Interface
Copy Paste
Realtime
Routers
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
(Select a Device to Drag and Drop to the Workspace)

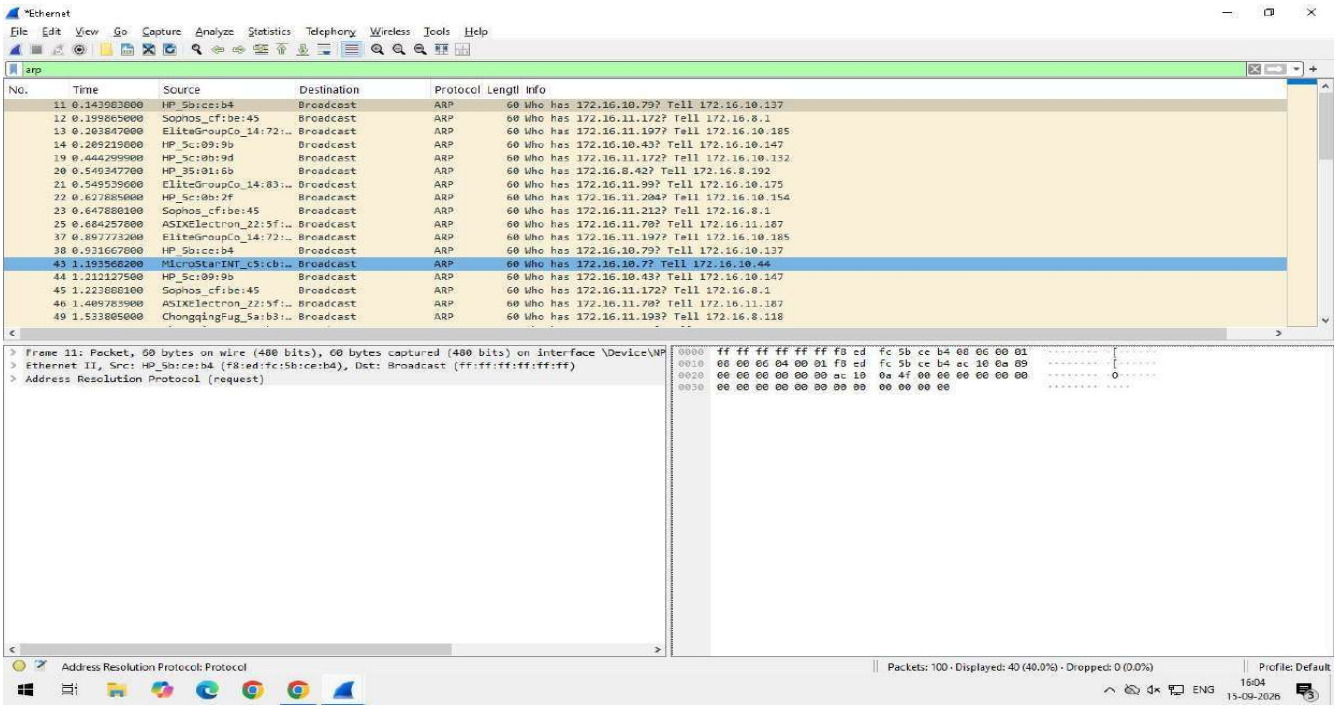
Experiment 5

1. Create a Filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

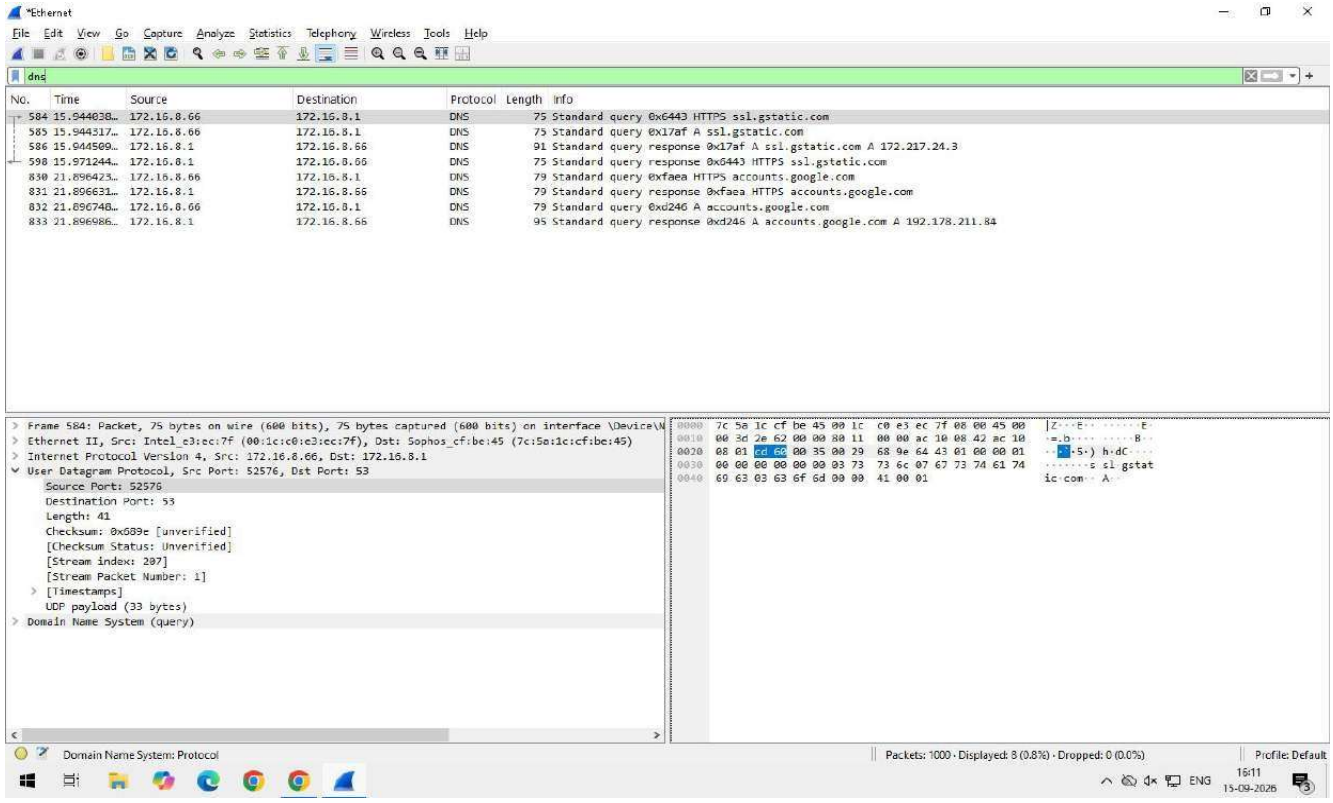


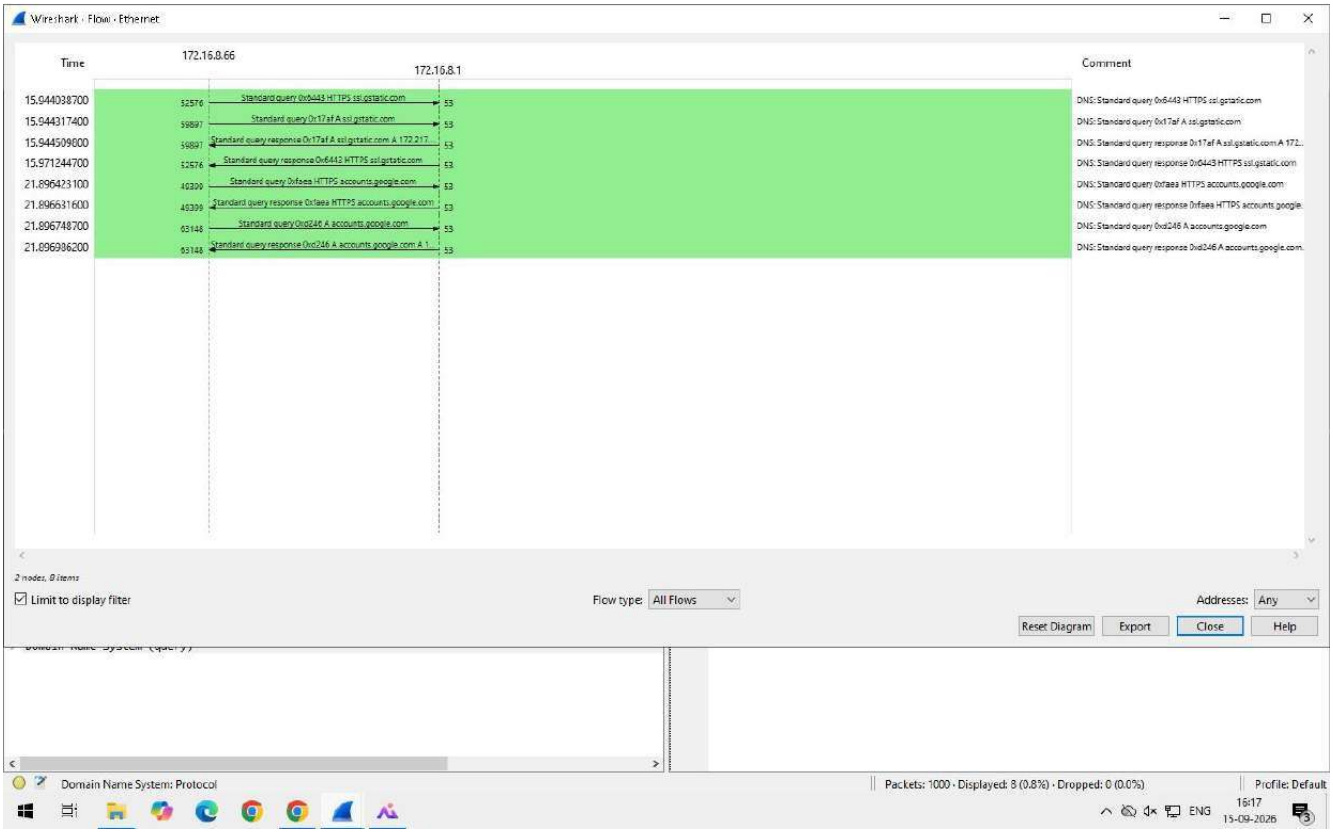


2. Create a Filter to display only ARP packets and inspect the packets.

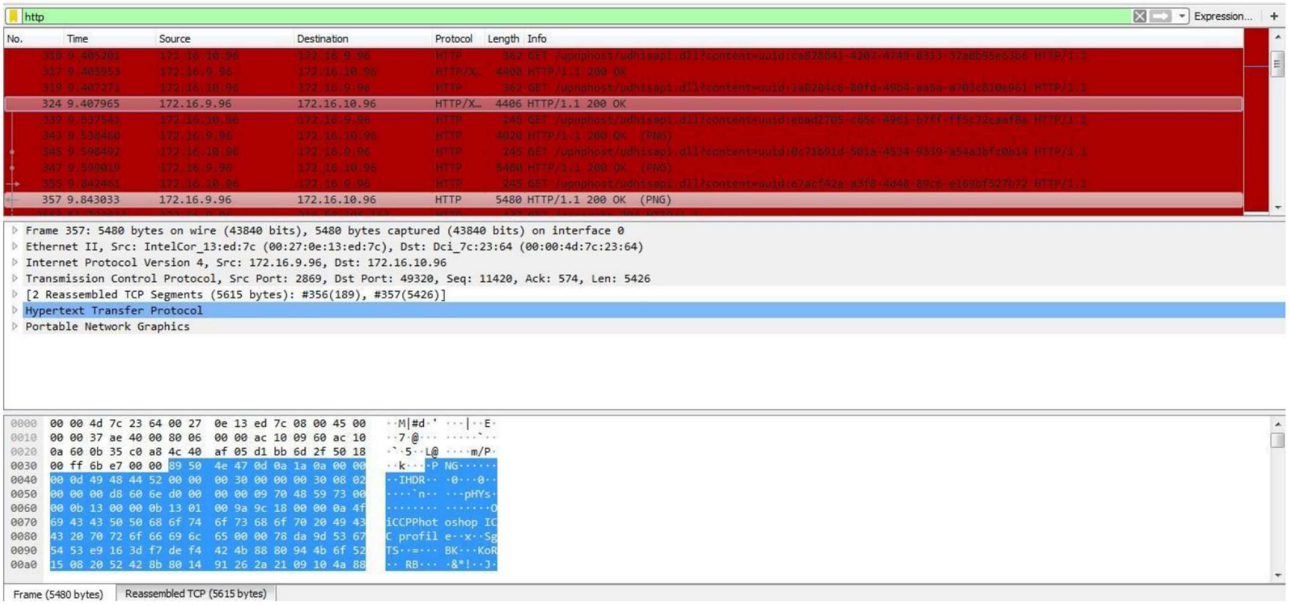


3. Create a Filter to display only DNS packets and provide the flow graph.

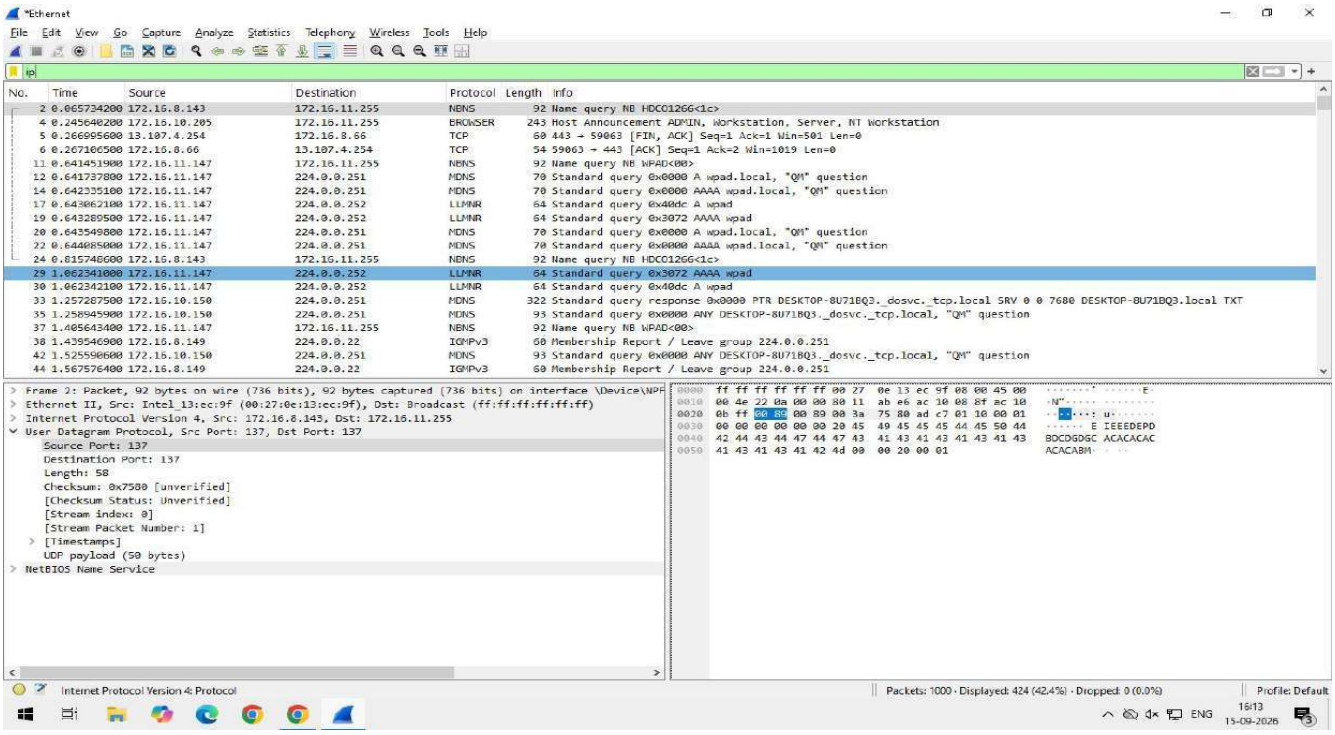




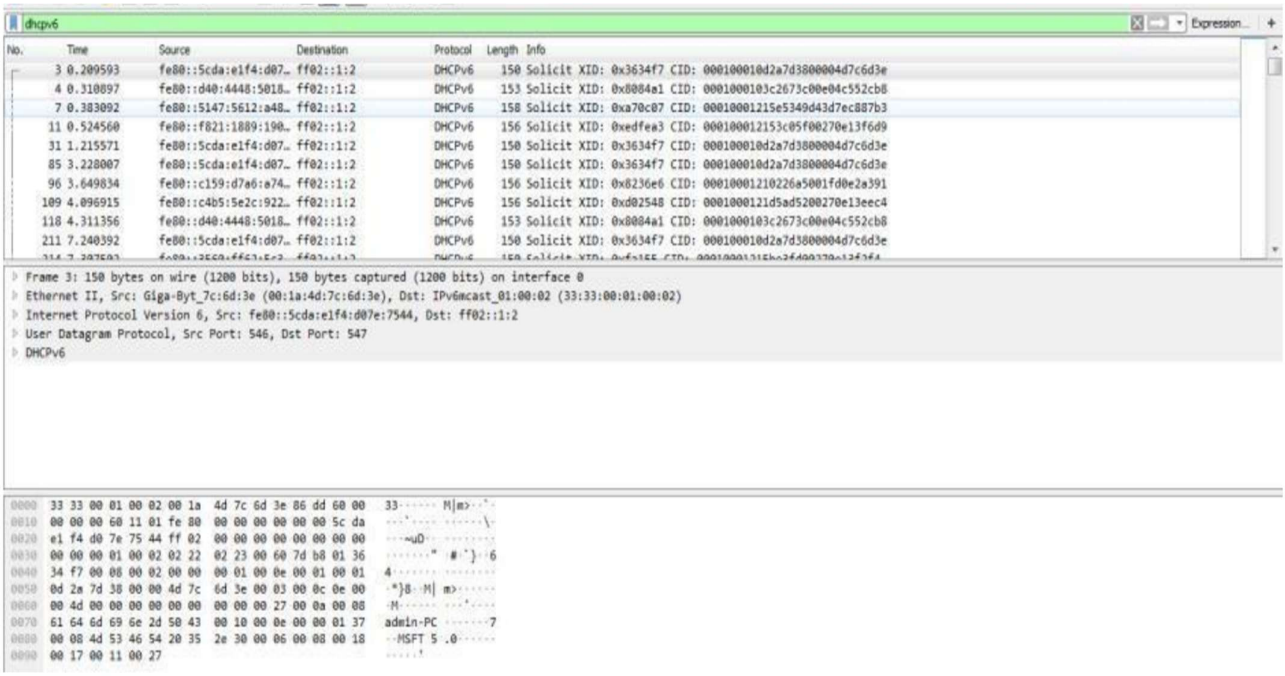
4. Create a Filter to display only HTTP packets and inspect the packets



5. Create a Filter to display only IP/ICMP packets and inspect the packets



6. Create a Filter to display only DHCP packets and inspect the packets.



Student Observation:

1. What is promiscuous mode?

Ans: A network interface mode that allows a device to capture and receive all network packets passing through the network, not just packets addressed to it.

2. Do ARP packets have transport layer header? Explain.

Ans: No. ARP operates between the Data Link and Network layers. ARP packets do not contain a TCP or UDP transport-layer header.

3. Which transport layer protocol is used by DNS?

Ans: DNS mainly uses UDP port 53. TCP port 53 is also used for zone transfers and large responses.

4. What is the port number used by http protocol?

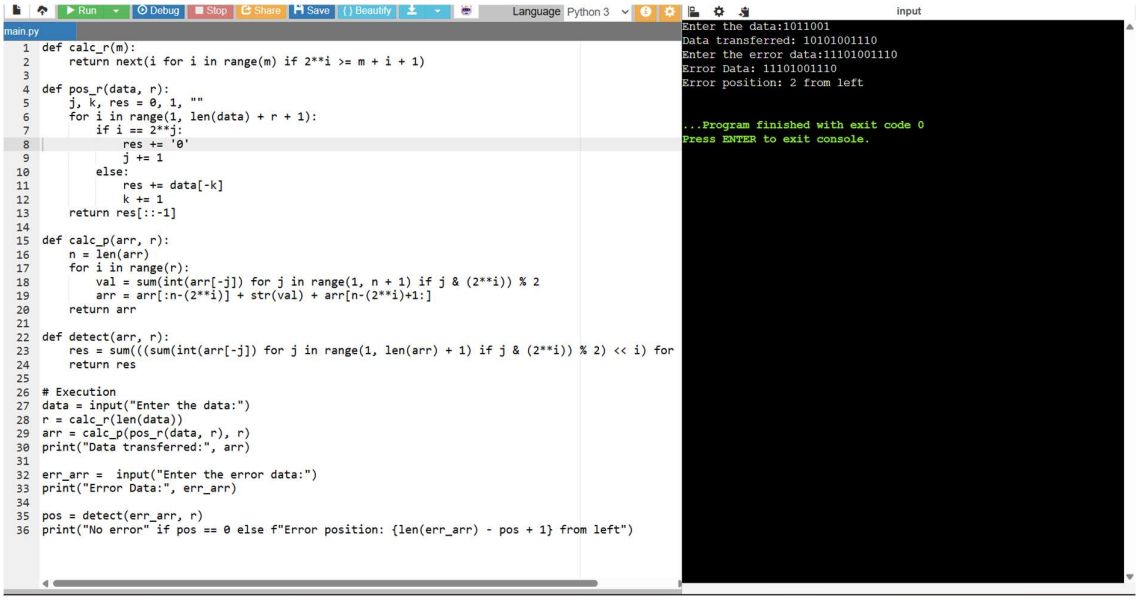
Ans: HTTP uses port 80 by default. HTTPS uses port 443.

5. What is a broadcast IP address?

Ans: An IP address used to send data to all devices in a network. For example, 192.168.1.255 is the broadcast address for a 192.168.1.0/24 network.

Experiment 6

Hamming Code



```
def calc_r(m):
    return next(i for i in range(m) if 2**i >= m + i + 1)

def pos_r(data, r):
    j, k, res = 0, 1, ""
    for i in range(1, len(data) + r + 1):
        if i == 2**j:
            res += '0'
            j += 1
        else:
            res += data[-k]
            k += 1
    return res[::-1]

def calc_p(arr, r):
    n = len(arr)
    for i in range(r):
        val = sum(int(arr[-j]) for j in range(1, n + 1) if j & (2**i)) % 2
        arr = arr[:n-(2**i)] + str(val) + arr[n-(2**i)+1:]
    return arr

def detect(arr, r):
    res = sum(((sum(int(arr[-j]) for j in range(1, len(arr) + 1) if j & (2**i)) % 2) << i) for
    return res

# Execution
data = input("Enter the data:")
r = calc_r(len(data))
arr = calc_p(pos_r(data, r), r)
print("Data transferred:", arr)

err_arr =  input("Enter the error data:")
print("Error Data:", err_arr)

pos = detect(err_arr, r)
print("No error" if pos == 0 else f"Error position: {len(err_arr) - pos + 1} from left")
```

```
Enter the data:1011001
Data transferred: 10101001110
Enter the error data:11101001110
Error Data: 11101001110
Error position: 2 from left

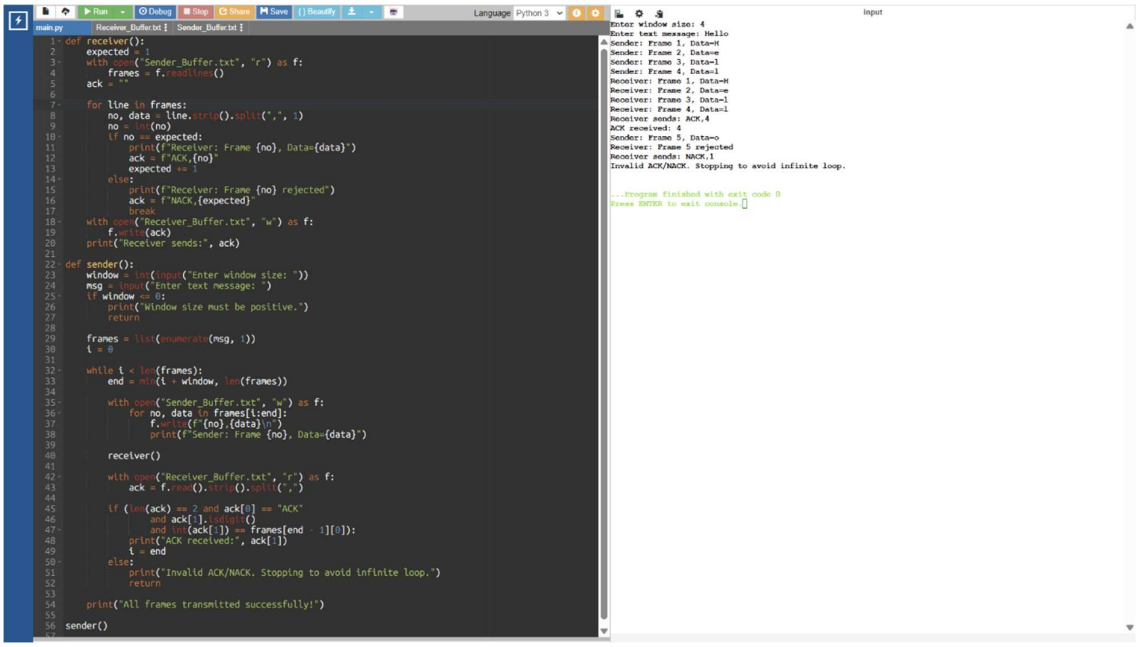

...Program finished with exit code 0
Press ENTER to exit console.
```

Result:

Given hamming code is successfully completed.

Experiment 7

Sliding Window



```
def receiver():
    expected = 1
    with open("Sender_Buffer.txt", "r") as f:
        frames = f.readlines()
    ack = ""

    for line in frames:
        no, data = line.strip().split(",", 1)
        no = int(no)
        if no == expected:
            print(f"Receiver: Frame {no}, Data={data}")
            ack = f"ACK,{no}"
            expected += 1
        else:
            print(f"Receiver: Frame {no} rejected")
            ack = f"NACK,{expected}"
            break
    with open("Receiver_Buffer.txt", "w") as f:
        f.write(ack)
    print("Receiver sends:", ack)

def sender():
    window = int(input("Enter window size: "))
    msg = input("Enter text message: ")
    if window <= 0:
        print("Window size must be positive.")
        return

    frames = list(enumerate(msg, 1))
    i = 0

    while i < len(frames):
        end = min(i + window, len(frames))

        with open("Sender_Buffer.txt", "w") as f:
            for no, data in frames[i:end]:
                f.write(f"{no},{data}\n")
                print(f"Sender: Frame {no}, Data={data}")

        receiver()

        with open("Receiver_Buffer.txt", "r") as f:
            ack = f.read().strip().split(",")

        if (len(ack) == 2 and ack[0] == "ACK"
                and ack[1].isdigit()
                and int(ack[1]) == frames[end - 1][0]):
            print("ACK received:", ack[1])
            i = end
        else:
            print("Invalid ACK/NACK. Stopping to avoid infinite loop.")
            return

    print("All frames transmitted successfully!")

sender()
```

```
Enter window size: 4
Enter text message: Hello
Sender: Frame 1, Data=H
Sender: Frame 2, Data=e
Sender: Frame 3, Data=l
Sender: Frame 4, Data=l
Receiver: Frame 1, Data=H
Receiver: Frame 2, Data=e
Receiver: Frame 3, Data=l
Receiver: Frame 4, Data=l
Receiver sends: ACK,4
ACK received: 4
Sender: Frame 5, Data=o
Receiver: Frame 5 rejected
Receiver sends: NACK,1
Invalid ACK/NACK. Stopping to avoid infinite loop.

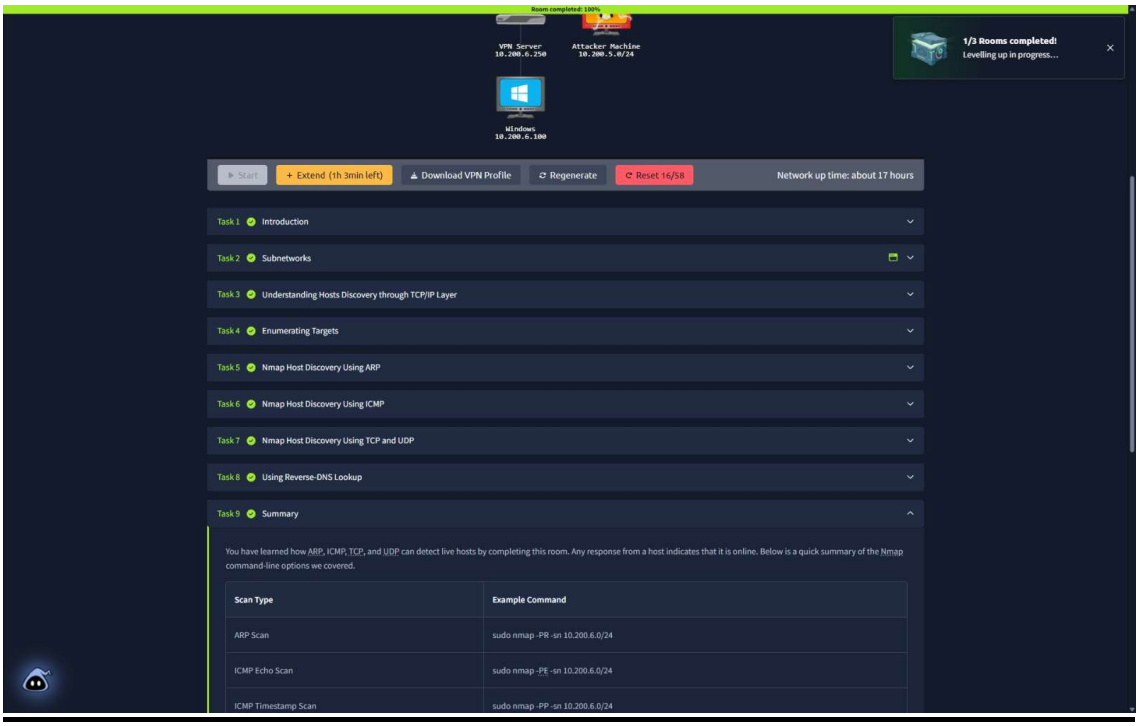
...Program finished with exit code 0
Press ENTER to exit console.
```

Result:

Given sliding window code is successfully completed.

Experiment 8

Nmap Using TryHackMe

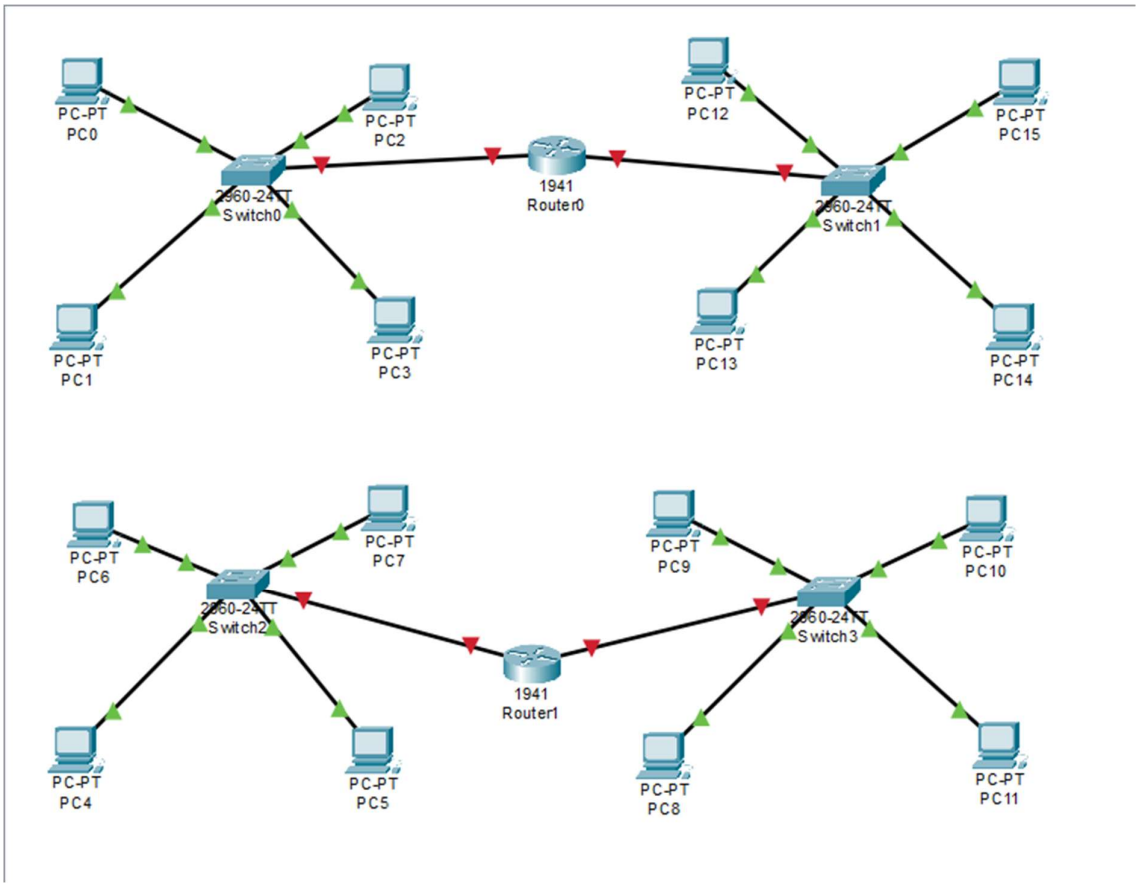


Result:

Given experiment is successfully completed.

Experiment 9

Subnetting



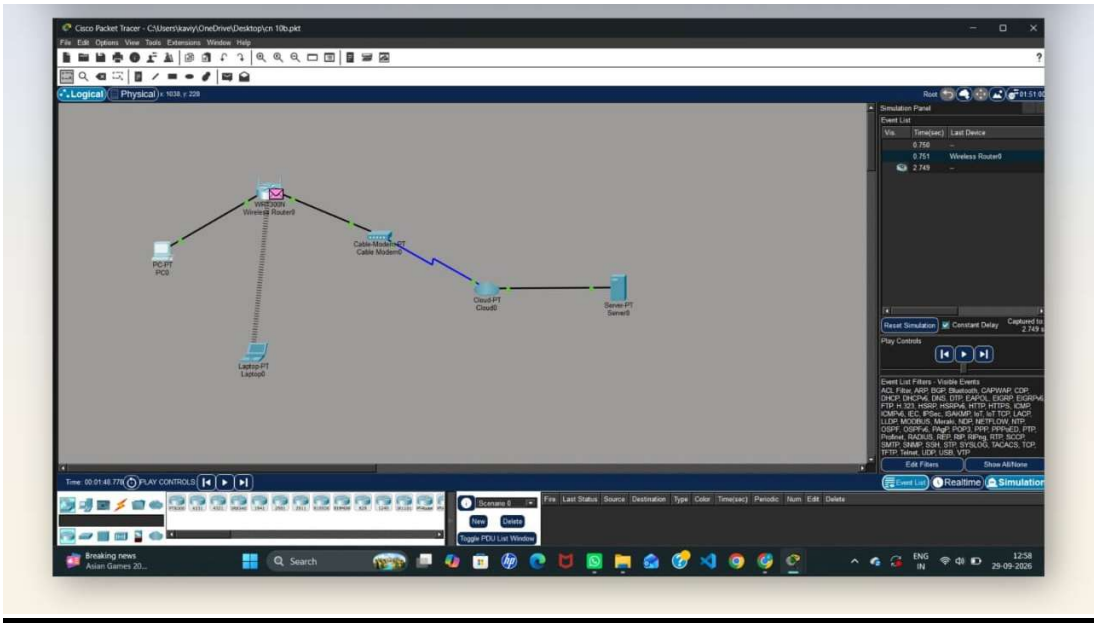
Result:

Given experiment is successfully completed.

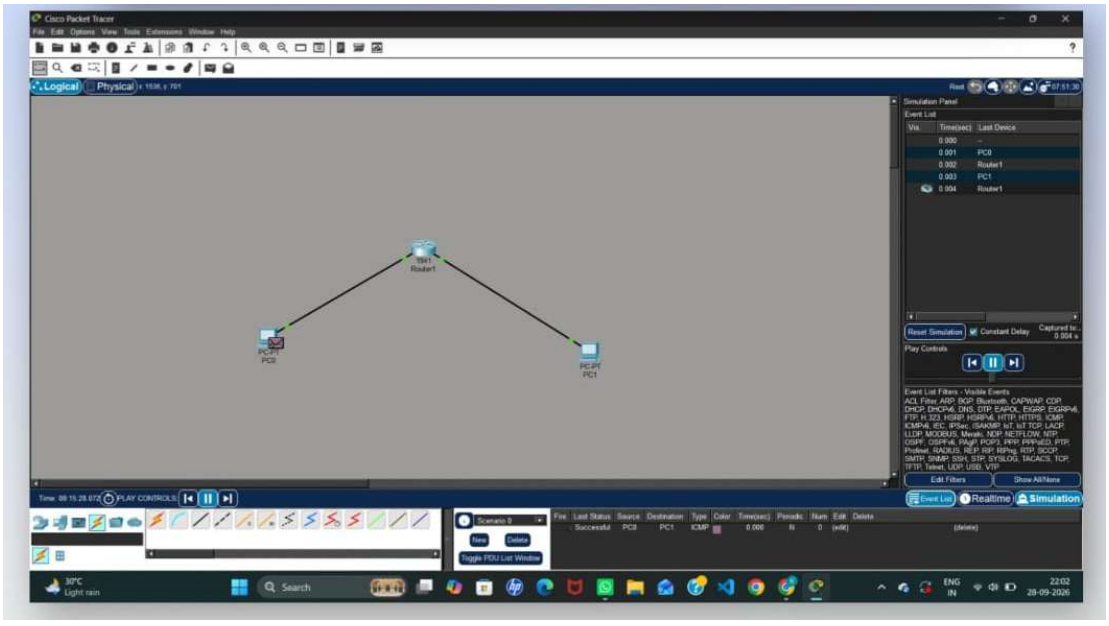
Experiment 10

Internetworking with Routers

a)



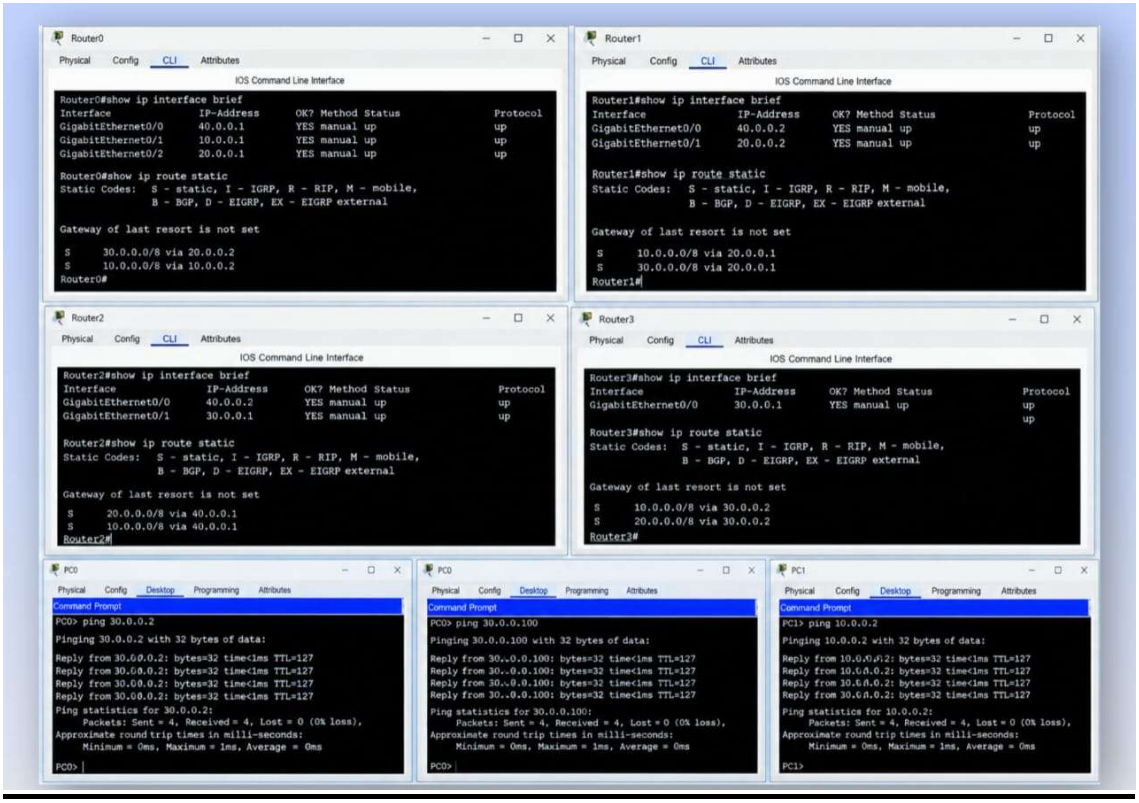
b)



Result: Given experiment is successfully completed.

Experiment 11

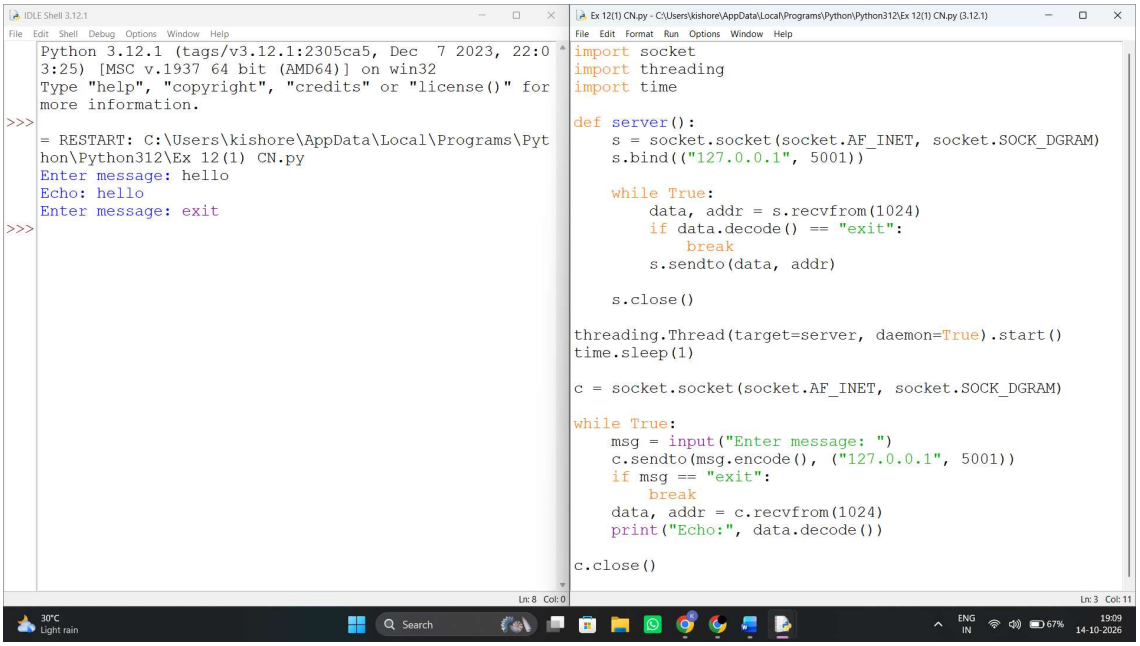
Static Routing Configuration



Result: Given experiment is successfully completed.

Experiment 12

Implement echo client server using TCP/UDP sockets.



```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading
import time

def server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(("127.0.0.1", 5001))

    while True:
        data, addr = s.recvfrom(1024)
        if data.decode() == "exit":
            break
        s.sendto(data, addr)

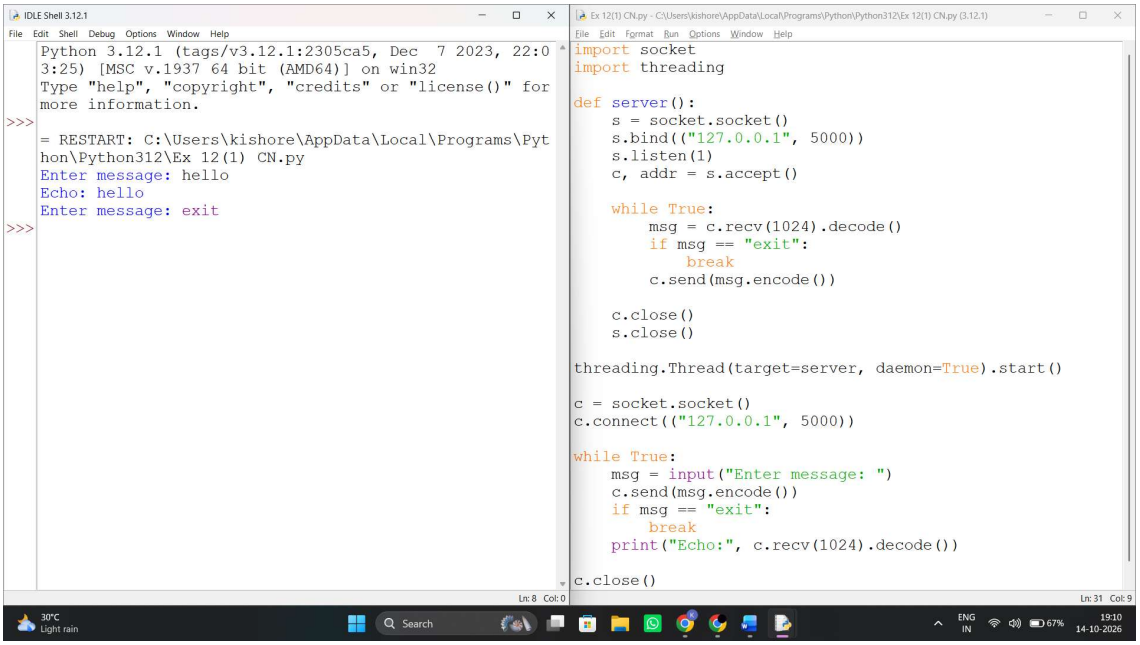
    s.close()

threading.Thread(target=server, daemon=True).start()
time.sleep(1)

c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

while True:
    msg = input("Enter message: ")
    c.sendto(msg.encode(), ("127.0.0.1", 5001))
    if msg == "exit":
        break
    data, addr = c.recvfrom(1024)
    print("Echo:", data.decode())

c.close()
```



```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec  7 2023, 22:03:25) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\kishore\AppData\Local\Programs\Python\Python312\Ex 12(1) CN.py
Enter message: hello
Echo: hello
Enter message: exit
>>>
```

```
import socket
import threading

def server():
    s = socket.socket()
    s.bind(("127.0.0.1", 5000))
    s.listen(1)
    c, addr = s.accept()

    while True:
        msg = c.recv(1024).decode()
        if msg == "exit":
            break
        c.send(msg.encode())

    c.close()
    s.close()

threading.Thread(target=server, daemon=True).start()

c = socket.socket()
c.connect(("127.0.0.1", 5000))

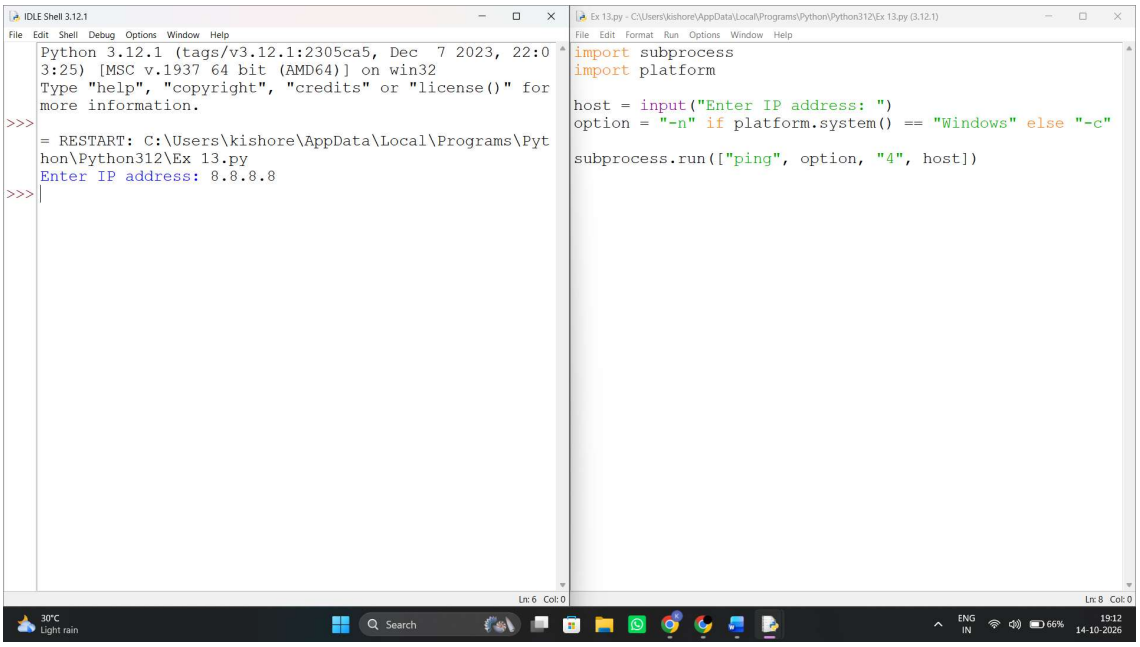
while True:
    msg = input("Enter message: ")
    c.send(msg.encode())
    if msg == "exit":
        break
    print("Echo:", c.recv(1024).decode())

c.close()
```

Result: Given experiment is successfully completed.

Experiment 13

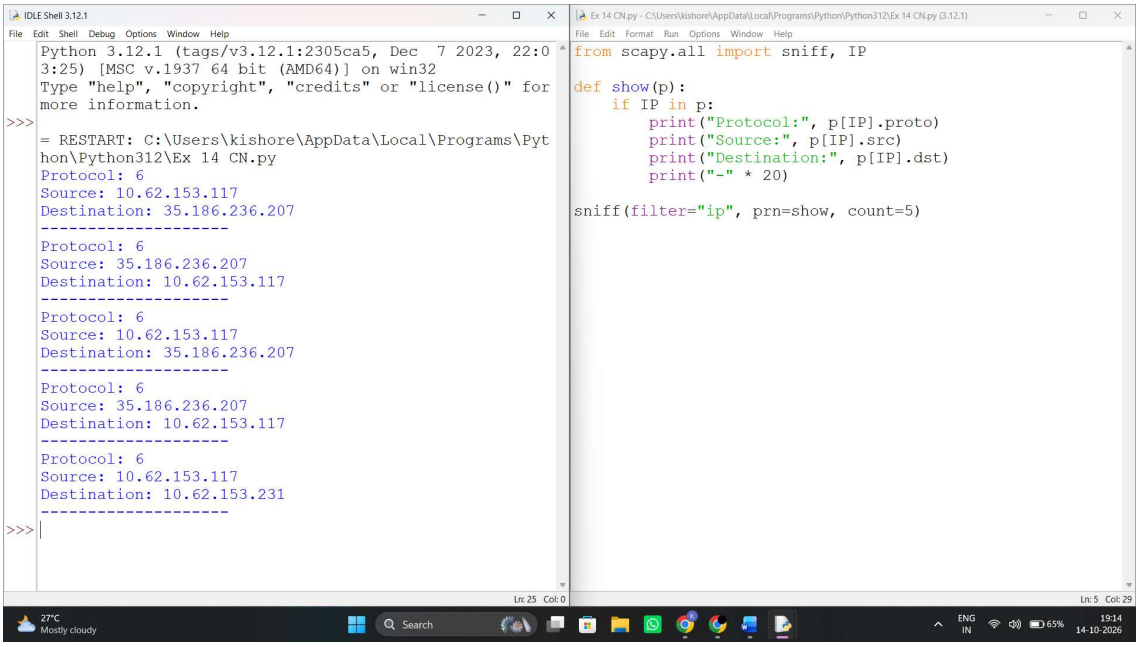
Implement your own ping program



Result: Given experiment is successfully completed.

Experiment 14

Write a code using RAW sockets to implement packet sniffing.



Result: Given experiment is successfully completed.